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Debate: Evidence-based AI risk assessment for public policy

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Outlooks on AI's risks are primarily influenced by proponents' worldviews—what they assume likely or desirable based on prior beliefs, rather than current technological capabilities and development trajectories. For regulating, funding and procuring AI, the public sector needs a more solid foundation of assessing AI than science fiction. This article provides a way forward from the point of view of an engineer and scientist, offering a pragmatic evaluation of 10 commonly cited AI risks and respective evidence-based recommendations for policy responses.

How worldviews shape AI discourse

AI as a technology and field, with more than 70 years of development history, is reasonably mature (Prince, 2023); the debate surrounding it isn't (Jordan, 2025). Virtually any voice weighing in on AI's risks does not argue based on what the technology is or isn't (Stadelmann, 2025), but what they want, or fear, it to be (Hao, 2025). In part, this is because achieving AI's formulated vision—*doing things with a computer that previously could only be done by humans with their minds*—included anthropomorphising the technology from the outset, which has led to inflated expectations and respective cycles of AI hypes and winters ever since (Segessenmann et al., 2025). But, much more, people resort to articulating their *beliefs* on issues where stakes are high, and while this is understandable and even commendable, these views should be subjected to scrutiny before informing public policy. An influential example is the TESCREAL bundle of philosophies (an acronym for transhumanism, extropianism, singularitarism etc.): identified as the driver behind much of the loudest voices in the AI field (Gebru & Torres, 2024), it suggests imminent existential risk that drove much of the first governmental reactions to AI but has since been called 'ridiculous' by a growing body of leading figures in the field (Heaven, 2023). Meanwhile, tech industry proponents used the ensuing FUD as a strategy that helped their bottom line. What are the implications of such reliance on worldviews for AI governance and public intervention, especially when it comes to assessing AI risks in order to regulate, fund, or procure AI? How could it be done evidence-based instead (van de Poel, 2016)?

A pragmatic, evidence-based approach to AI risk assessment

Many lists of AI risks exist in the literature, with striking similarities; for example, Jackson (2024) lists 10 risks of AI. In this article I give a brief assessment of each risk with respect to public policy, based on the view that AI is, basically, a normal technology (Narayanan & Kapoor, 2025) and should be treated as such—especially in the absence of any hard evidence to the contrary (Kambhampati et al., 2025; Kumar et al., 2025).

Existential risk

Alleged existential threats are purely based on the assumption of imminent 'artificial general/superintelligence'. No plausible scientific path to its creation exists, and the underlying worldview-based predictions arguably get most of their attention not because of their likelihood, but because of the visibility and economic power of their proponents. Hence, I assess these risks as purely hypothetical; they must not drive policy.

Dependence

I see our own human dependence on AI systems as a major risk: giving away freedom of choice and decision-making voluntarily to assumed machine competence as well as sacrificing personal growth in skills and also character through AI overuse hurt human autonomy. Strengthening moral formation through making people aware of their own value and dignity is a reasonable counter measure (Stadelmann, 2025). This needs to be reflected, for instance, in school curricula and societal institutions: AI is a prime occasion to rethink what (and what for) the human is (Segessenmann et al., 2025), and what we want to keep as inherently human must also guide public funding and procurement. As AI predictions lead to suggestions that ultimately shape human behaviour (de Cárdenas, 2025), dependency's further consequences of, for example, economic and political nature, finally, need regulatory attention.

A power monopoly

The economics of AI, foremost its business models and ways of distribution, should attract more attention from a regulatory and taxation perspective. As with social media (Orlowski, 2020), the specific way business is conducted in the contemporary tech industry (especially the proclaimed 'end game' of establishing a monopoly and the extreme high stakes in spending) holds much potential for harm

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(Zitron, 2025) and needs to be observed and suitably regulated to avoid similarly bad side effects. Tech sovereignty, i.e. the fundamental freedom of choice of domestic or open source alternatives along the whole AI system value chain, needs to be achieved, which is a task for public funding and procurement.

Ethical dilemmas

Ethical dilemmas are numerous. For instance, the practice of aligning AI systems with human preferences as an AI safety measure produces 'sycophantic' AI systems, making general-purpose chatbots a catastrophic counsellor for serious mental health issues. For the sake of society and the economy, governments need to invest into *prohuman* AI design methodologies and use cases, where the affordances a system offers are morally aligned with what is good for a human being (Schirch, 2023; Acemoğlu, 2025). This should be a funding priority and procurement requirement.

Misinformation

It is best to assume that AI generated content is already unidentifiable with any reasonable confidence. I suggest, however, that people adapt socially: not liking the feeling of being fooled, we will shift our trust back into smaller, more local, more personal relationships. This makes misinformation a lesser public concern apart from criminal offences.

Job displacement

Many opinions on AI's effect on the labour market are based on the unrealistic assumption of artificial general intelligence —AGI (Narayanan & Kapoor, 2025). Instead, AI does not automate jobs, but tasks, and cannot even come close to automating the 'glue' between tasks that makes up most of the complexity in human business. I assume a rather manageable impact on jobs on the societal level but shifts and changes playing out on the individual level. Hence, government can support this in driving curricula in (continuing) education and fostering an evidence-based discourse on the expected change to counter unwarranted fears. Digitally mature citizens are a governmental task, achieved by education. This maturity also helps organizations to make better decisions on what can and should be automated (Ivanova, 2025).

Security threats

AI is a dual-use technology that will be used to wilfully do harm. I see an arms race between illegal forces and law enforcement that currently seems reasonably balanced. Governments need to continually support public security institutions to keep pace without drifting into surveillance and authoritarianism. Security threats through not having technology sovereignty have to be addressed (see above).

Data privacy concerns

Existing regulation, for example in the EU, seems adequate in principle but currently has a hard stand against an industry that largely ignores certain law and user preferences for the promise of profit (Crofts, 2024). What is and isn't fair use needs to be carefully reconsidered, having not just the most heavily venture-capital-backed stakeholders in mind.

Lack of transparency

Depending on the AI methods used, results are not directly comprehensible for humans (as is also the case with the decisions of fellow humans, and similarly, not every AI result needs an explanation). The field of explainable AI (Bennetot et al., 2024) and new approaches to building machine learning systems (von der Malsburg, 2022; Kumar et al., 2025) offer remedies in the cases where explainability is a reasonable requirement. Regulatory requirements should be appropriate to the context and stakeholders (for example not stifling much-needed open source developments), and more R&D effort should be directed into the development of respective methods through public funding.

Bias and discrimination

As far as 'algorithmic bias' is concerned, the problem is well understood as rooted in the data (Wehrli, 2022) and finds adequate technical solutions for example in the field and tools of algorithmic fairness (Hertweck, 2023), as well as in existing regulation on non-discrimination. This is no longer an issue of regulation or debate, but of proper implementation (which presents its own challenges).

Recommendations for public policy

According to the assessment above, few risks associated with AI adoption can be considered solved, but all risks appear mitigatable: short-term risks through technical, social, or by regulatory means as pointed out above and in the literature; longer-term risks by monitoring appropriate evidence. As long as they appear hypothetical, such 'risks' should not influence practice (for example policy-making, organizational governance, or the societal debate). The three most prominent recommendations to be gleaned from this article are:

- *Invest* in education on what AI is and isn't; it is the number one government intervention against the major risks of AI dependence and widespread anxiety, with their politically and economically destabilizing effects.
- *Regulate* AI business models with the goal of creating tech sovereignty, also through public procurement and open source; these are the most necessary interventions to sustain a fair market and equitable societies where free choice persists.
- *Fund* innovation in the direction of less energy and data-hungry, more common-sensical and fundamentally prohuman AI, which is to be found off the currently beaten path of scaling large language models (LLMs); this is the greatest service to higher degrees of transparency, fewer ethical dilemmas and better privacy, and should also include incentives for the private sector to invest and build.

My main takeaway for policy-makers: don't fear hypothetical risks, but act according to the evidence.

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